XIAO-YU GUO

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RESEARCH PROFILE

AI researcher with a PhD in Artificial Intelligence from Monash University and research experience spanning academia and industry. My research focuses on natural language processing, large language models, complex reasoning, multimodal learning, and responsible AI, with publications in leading international conferences, including EMNLP and COLING. I have a strong track record of developing novel machine learning methods, constructing evaluation benchmarks, and investigating model robustness, fairness, and generalisation. My experience includes conducting independent and collaborative research, supervising research students, and translating research outcomes into practical AI applications. I am particularly interested in advancing trustworthy, efficient, and generalisable AI systems.

PROFESSIONAL EXPERIENCE

Oracle Corporation Australia Pty Limited *Melbourne, VIC*
Senior Applied Scientist *Jan 2025 ~ May 2026*

Applied science role spanning real-world data handling, ambiguous problem solving, medical-domain agent workflows, model evaluation and validation in a customer-facing environment.

- Real-world data handling: consolidated domain knowledge from customers and subject-matter experts, and built automated pipelines for real-world data selection, ingestion, preprocessing and post-processing to support downstream AI workflows.
- Ambiguous problem solving worked closely with stakeholders to clarify ambiguous requirements, designed scalable AI solutions that translated evolving requirements into deployable systems.
- Agent workflow design: designed and implemented agent workflows within the medical domain to support domain-specific use cases, integrating LLM-based automation to improve task execution and usability.
- Model evaluation and validation: conducted evaluation experiments, including performance testing, validating, and robustness assessment of deployed models to ensure reliability and effectiveness in real-world settings.

Australian Institute for Machine Learning *Adelaide, SA*
Postdoctoral Research Fellow *Nov 2023 ~ Jan 2025*

Research role focused on machine learning, system robustness, responsible AI, algorithm design, model optimization, technical documentation and publication outputs.

- Machine learning research: conducted research on fundamental machine learning challenges, with a particular focus on robustness, reliability and responsible AI.
- Algorithm design and evaluation: analyzed system requirements, designed novel algorithms, and evaluated their effectiveness through empirical experimentation using quantitative metrics.
- Model optimization: collaborated closely with colleagues to refine technical specifications, investigate system limitations and improve model performance.
- Research communication: contributed to technical documentation and research publications, and presented findings, methodologies and results to the research community.

EDUCATION & CREDENTIALS

FIT / Monash University *Melbourne, VIC*
Doctor of Philosophy, Artificial Intelligence *Mar 2020 ~ Apr 2024*
Full Scholarship Recipient

Thesis: *Complex Question Answering: from Data to Method*

Research focus: designed and implemented machine learning models, including Neural Module Networks, and LLM-based Single- and Multi-modal architectures, to address complex question answering tasks under real-world data bias and distribution challenges.

SCSE / Beihang University *Haidian, Beijing*
Master of Engineering, Computer Science and Technology *Sep 2016 ~ Jan 2019*

Honors & awards: Outstanding Postgraduate Student in Beijing (2019); Outstanding Postgraduate Student, Beihang University (2019).

SCSE / Beihang University *Haidian, Beijing*
Bachelor of Engineering *Sep 2012 ~ Jun 2016*

SELECTED PUBLICATIONS

First Author (Question Answering, Natural Language Processing, Relation Classification)

- Xiao-Yu Guo, Yuan-Fang Li, and Gholamreza Haffari. 2023. DeSIQ: Towards an Unbiased, Challenging Benchmark for Social Intelligence Understanding. In Proceedings of EMNLP 2023.
- Xiao-Yu Guo, Yuan-Fang Li, and Gholamreza Haffari. Complex Reading Comprehension Through Question Decomposition. In Proceedings of the 20th ALTA, 2022.
- Xiao-Yu Guo, Yuan-Fang Li, and Gholamreza Haffari. Improving Numerical Reasoning Skills in the Modular Approach for Complex Question Answering on Text. In Findings of the EMNLP 2021.
- Xiaoyu Guo, Yuan-Fang Li, and Gholamreza Haffari. Understanding Unnatural Questions Improves Reasoning over Text. In Proceedings of the 28th COLING, 2020.
- Xiaoyu Guo, Meng Chen, Yang Song, Xiaodong He, and Bowen Zhou. Automated Thematic and Emotional Modern Chinese Poetry Composition. In Proceedings of the 8th NLPCC 2019.
- Xiaoyu Guo, Hui Zhang, Haijun Yang, Lianyuan Xu, and Zhiwen Ye, A Single Attention-Based Combination of CNN and RNN for Relation Classification. In IEEE Access, 2019.

Others (Question Answering, Natural Language Generation, Entity Linking)

- Afshar Shamsi, Xiao-Yu Guo, Hamid Alinejad-Rokny, Arash Mohammadi, Damien Teney, and Ehsan Abbasnejad. Multi-hypothesis Test-Time Adaptation to Mitigate Underspecification. In ECCV 2026.
- Bao Gia Doan, Afshar Shamsi, Xiao-Yu Guo, Arash Mohammadi, Hamid Alinejad-Rokny, Dino Sejdinovic, Damien Teney, Damith C Ranasinghe, and Ehsan Abbasnejad. Bayesian low-rank learning (Bella): A practical approach to Bayesian neural networks. In Proceedings of AAAI 2025.
- Peng Chen, Xiao-Yu Guo, Yuan-Fang Li, Xiaowang Zhang, Zhiyong Feng. Mitigating Language Bias of LMMs in Social Intelligence Understanding with Virtual Counterfactual Calibration. In Proceedings of EMNLP 2024.
- Fatemeh Shiri, Xiao-Yu Guo, Mona Golestan Far, Xin Yu, Reza Haf, Yuan-Fang Li. An Empirical Analysis on Spatial Reasoning Capabilities of Large Multimodal Models. In Proceedings of EMNLP 2024.
- Jiayi Chen, Xiao-Yu Guo, Yuan-Fang Li, and Gholamreza Haffari. Teaching Neural Module Networks to Do Arithmetic. In Proceedings of the 29th COLING, 2021.
- Lei Shen, Xiaoyu Guo, and Meng Chen. Compose like humans: Jointly improving the coherence and novelty for modern Chinese poetry generation. In 2020 IJCNN.

SELECTED EARLIER RESEARCH EXPERIENCE

FIT / Monash University *Melbourne, VIC*
Research Assistant on Computational Cultural Understanding *Feb 2022 ~ Jul 2023*

- Norm detection and discovery: without labeled data, built a model to classify social norms (like greetings, apology, etc.) in Chinese culture, using multimodal inputs including videos, audios and transcripts.
- Social intelligence question answering: based on the Social-IQ benchmarks, focused on fusion and reasoning over multimodal data, proposed a methodology on dataset bias detection and debiasing, and got one academic paper accepted in EMNLP 2023.

NLG Group / JD AI *Chaoyang, Beijing*
Internship on Chinese Poetry Generation *Jul 2018 ~ May 2019*

- Modern Chinese poetry generation: started from a basic seq2seq+attention model, incorporated human emotions, topics and carried out experiments on millions of datapoints to achieve more coherent, expressive, and thematic poem generation.
- Modern Chinese poetry refinement: a follow-up work of poetry generation, generate-then-refine poems according to human minds by adding encoders encoding useful poem lines.
- Knowledge driven dialogue system: leveraged information in knowledge graphs, use OpenNMT to generate multi-turn dialogue between machines and humans.

State Key Laboratory of SDE / Beihang University *Haidian, Beijing*
Research Assistant on Relation Classification *Mar 2018 ~ Jan 2019*

- Relation Classification (RC): built a novel RC model using the combination of RNN and CNN, conducted experiments on SemEval-2010 task 8 dataset and got the highest F1 score compared to previous SOTA models.
- Refined RC model: a follow-up work of the relation classification model, slightly modified and extended the datasets (SemEval-2018 task 7 and KBP37) and published a paper in IEEE Access.

Scholarship & Honor

Outstanding Postgraduate Student in Beijing	Jan. 2019
Outstanding Postgraduate Student in Beihang University (20 out of 235)	Mar. 2018
Second Price Scholarship in Beihang University	Oct. 2017
Outstanding Undergraduate Student in Beihang University	Jun. 2016
Outstanding Student Leader in Beihang University	Oct. 2015
Learning Outstanding Scholarship in Beihang University	Dec. 2014

Competition Reward

“First Price” in Asia Student Supercomputer Challenge, 2015	May 2015
CCF Score: 300, Ranking: 5.73%	Nov. 2014

Association Participation & Volunteering

SCSE 120615 class monitor in Beihang University	Sep. 2014 ~ Jul. 2015
INCLUDED volunteer (Service Dates: July 14~July 25)	Jul. 2014